

PHY201: Waves & Optics

Problem Set 8

November 3, 2025

1. Write down the electric and magnetic fields for a monochromatic plane wave of amplitude E_0 and frequency ω which is
 - (a) Traveling in the negative x direction and polarized in the z direction.
 - (b) Traveling in the direction from origin to the point $(1, 1, 1)$, with polarization parallel to the xz plane. In each case give explicit Cartesian components of $\mathbf{k}$ and $\hat{\mathbf{n}}$.
2. In the derivation of Fresnel's equations, we assumed that the polarisation of the incident, reflected, and transmitted waves must be in the same plane. Assume that the polarisation vectors for the reflected and transmitted waves are in the direction $\hat{\mathbf{n}}_R = \cos \theta_R \hat{\mathbf{x}} + \sin \theta_R \hat{\mathbf{y}}$ and $\hat{\mathbf{n}}_T = \cos \theta_T \hat{\mathbf{x}} + \sin \theta_T \hat{\mathbf{y}}$, and use the boundary conditions to show that $\theta_R = \theta_T = 0$ (Polarisation vector for the incident wave is in the xz plane, as we assumed in the derivation done in the class.)
3. For oblique incidence and with polarization of the incident wave parallel to the plane of incidence, express the Fresnel coefficients $r_{\parallel} = E_{0R}/E_{0I}$ and $t_{\parallel} = E_{0T}/E_{0I}$ in terms of the angles θ_I and θ_T . Assume non-magnetic media, i.e. $\mu_1 \approx \mu_2 \sim \mu_0$. The reflection and transmission coefficient for a monochromatic plane wave incident at an interface between two media are defined by $R = I_R/I_I$ and $T = I_T/I_I$, respectively. In this definition, the intensities of the incident (I_I), reflected (I_R), and transmitted (I_T), waves are defined as the component normal to the interface of the average (over one period) of the Poynting vectors (e.g., $I_I = \langle \mathbf{S}_I \rangle \cdot \hat{\mathbf{z}}$). Show that $R + T = 1$.
 - (a) Show that there is no reflection when $n_2 = n_1$.
 - (b) If the angle of incidence is such that $\theta_I + \theta_R = \pi/2$, then there is no reflected beam. This in turn implies that if a beam is incident at an angle such that $\theta_I + \theta_R = \pi/2$, then the parallel component of the $\mathbf{E}$ vector will not be reflected and the reflected light will be polarized with its $\mathbf{E}$ vector perpendicular to the plane of incidence. This is the famous Brewster law. The corresponding angle of incidence is known as the Brewster angle (or the polarizing angle) and is usually denoted by θ_B . Show that

$$\theta_B = \tan^{-1} \left(\frac{n_2}{n_1} \right).$$

- (c) When light is incident on a denser medium, there is a phase change of π . If the media are interchanged, the angles of incidence and refraction are reversed. In other words, if $r'_{\parallel}$ and $t'_{\parallel}$ denote the Fresnel coefficients, then show that the following Stokes equation will hold:

$$1 + r_{\parallel} r'_{\parallel} = t_{\parallel} t'_{\parallel}.$$

- (d) At grazing incidence, there is total reflection.
- (e) Show that when an e-m wave is incident on a rarer medium (i.e. $n_2 < n_1$), then for $\theta_I > \theta_C$, there is total internal reflection.

4. (a) Show that the skin depth in a poor conductor ($\sigma \ll \omega\epsilon$) is

$$\delta = \frac{2}{\sigma} \sqrt{\frac{\epsilon}{\mu}}$$

(independent of frequency). Find the skin depth (in meters) for (pure) water. (Use the static values of ϵ , μ , and σ ; your answers will be valid, then, only at relatively low frequencies.)

- (b) Show that the skin depth in a good conductor ($\sigma \gg \omega\epsilon$) is

$$\delta = \frac{\lambda}{2\pi}$$

(where λ is the wavelength in the conductor). Find the skin depth (in nanometers) for a typical metal ($\sigma \approx 10^7 (\text{m}\Omega)^{-1}$) in the visible range ($\omega \approx 10^{15} \text{ s}^{-1}$), assuming $\epsilon \approx \epsilon_0$ and $\mu \approx \mu_0$. Why are metals opaque?

- (c) Show that in a good conductor the magnetic field lags the electric field by 45° , and find the ratio of their amplitudes. For a numerical example, use the “typical metal” in part (b).